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Pedal processing for primary / alternate aircraft braking system architecture

Abstract

An aircraft braking system is disclosed that includes a first pedal, first and second pedal sensors for the first pedal, a first brake control unit (e.g., primary), and a separate second brake control unit (e.g., secondary). The first pedal position sensor is operatively interconnected with one of the first brake control unit and the second brake control unit, while the second pedal position sensor is operatively interconnected with the other of the first brake control unit and the second brake control unit. Outputs from these pedal sensors may be used to control operation of the aircraft braking system in at least some fashion.

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Background/Summary

FIELD

(1) The present disclosure generally relates to the field of aircraft and, more particularly, to control of aircraft braking systems.

BACKGROUND

(2) An aircraft braking system may utilize multiple brake control units, including where both the

pilot and co-pilot each have a set of pedals to control the aircraft braking system. Multiple brake control units and multiple sets of pedals each may be utilized for redundancy, but can also result in a certain degree of complexity. In any case and regardless of complexities, an aircraft braking system should be configured to reduce the potential for un-commanded braking during at least certain conditions, such as during takeoff.

SUMMARY

(3) An aircraft braking system is presented herein. Both the configuration of such an aircraft braking system and the operational characteristics/operation of such an aircraft braking system are within the scope of this Summary. The aircraft braking system includes at least one pedal (e.g., a first pedal), a first pedal position sensor for this first pedal, a second pedal position sensor for this first pedal, a first brake control unit, and a second brake control unit. The first pedal position sensor is operatively interconnected with one of the first brake control unit and the second brake control unit (i.e., the first pedal position sensor is not operatively connected with both the first brake control unit and the second brake control unit), while the second pedal position sensor is operatively interconnected with the other of the first brake control unit and the second brake control unit (i.e., the second pedal position sensor is not operatively connected with both the first brake control unit and the second brake control unit). A comparison module (e.g., in software or hardware) is configured to compare a first output with a second output. The first output is provided by the first brake control unit from an output of the first pedal position sensor and that relates to a first pedal position of the first pedal. The second output is provided by the second brake control unit from an output of the second pedal position sensor and that relates to a second pedal position. A first condition precludes operation of the aircraft braking system in response to a displacement of the first pedal. This first condition exists when: 1) the comparison module receives the first output from the first brake control unit; 2) the comparison module receives the second output from the second brake control unit; and 3) the comparison module determines that the first output and the second output fail to comply with a first predetermined standard (e.g., the first output and the second output do not adequately match one another or are not within a predetermined threshold of one another in one or more aspects).

(4) Various aspects of the present disclosure are also addressed by the following examples and in the noted combinations: 1. An aircraft braking system comprising: a first pedal pair comprising a first pedal and a second pedal; a first pedal position sensor for said first pedal; a second pedal position sensor for said first pedal; a first brake control unit; a second brake control unit, wherein said first pedal position sensor is operatively interconnected with one of said first brake control unit and said second brake control unit and said second pedal position sensor is operatively interconnected with the other of said first brake control unit and said second brake control unit; and a first comparison module configured to compare a first output with a second output, wherein said first output is provided by said first brake control unit from an output of said first pedal position sensor and relates to a first pedal position of said first pedal, wherein said second output is provided by said second brake control unit from an output of said second pedal position sensor and relates to a second pedal position, and wherein said first brake control unit comprises said first comparison module; wherein a first condition precludes operation of said aircraft braking system in response to displacement of said first pedal, said first condition comprising: 1) said first comparison module receiving said first output from said first brake control unit and receiving said second output from said second brake control unit; and 2) said first comparison module determining that said first output and said second output fail to comply with a first predetermined standard. 2. The aircraft braking system of paragraph 1, wherein said first output comprises both a validity and magnitude relating to said first pedal position, and wherein said second output comprises both a validity and magnitude of said second pedal position. 3. The aircraft braking system of any of paragraphs 1-2, wherein one of said first brake control unit and said second brake control unit is a primary brake control unit and the other of said first brake control unit and said

second brake control unit is a secondary brake control unit. 4. The aircraft braking system of any of paragraphs 1-3, wherein said first brake control unit comprises a first outboard brake control card (BCC) and a first inboard BCC, wherein said second brake control unit comprises a second outboard BCC and a second inboard BCC, and wherein said first inboard BCC of said first brake control unit comprises said first comparison module. 5. The aircraft braking system of paragraph 4, wherein said first pedal position sensor is operatively interconnected with said first inboard BCC of said first brake control unit and said second pedal position sensor is operatively interconnected with said second outboard BCC of said second brake control unit. 6. The aircraft braking system of paragraph 5, wherein said first output is provided by said first inboard BCC of said first brake control unit and said second output is provided by said second outboard BCC of said second brake control unit. 7. The aircraft braking system of any of paragraphs 5-6: wherein said first inboard BCC of said first brake control unit comprises said first comparison module, a first module, and a second module; wherein said first module utilizes a third output from said first comparison module and a fourth output, wherein said fourth output is provided by said first brake control unit from an output of said first pedal position sensor and relates to said first pedal position of said first pedal; and wherein said first module outputs a fifth output to said second module. 8. The aircraft braking system of paragraph 7, wherein a sixth output is provided by said first brake control unit to said second module from an output of said first pedal position sensor and that relates to a third pedal position of said first pedal, and wherein said second module is operatively interconnected with a first shut-off valve driver that opens a first shut-off valve of said aircraft braking system when said fifth output satisfies a first condition and said sixth output satisfies a second condition. 9. The aircraft braking system of paragraph 8, wherein satisfaction of said first condition at least requires said first pedal position to be both above a first predetermined threshold and to be valid, and wherein satisfaction of said second condition at least requires said third pedal position to be above a second predetermined threshold. 10. The aircraft braking system of any of paragraphs 8-9: wherein said first outboard BCC of said first brake control unit comprises a third module; wherein said first module also transmits said fifth output to said third module; wherein said first inboard BCC of said first brake control unit also transmits said sixth output to said third module; and wherein said third module is operatively interconnected with a second shut-off valve driver that opens a second shut-off valve of said aircraft braking system when said fifth output satisfies said first condition and said sixth output satisfies said second condition. 11. The aircraft braking system of paragraph 10, wherein said first module transmits said fifth output to said third module via an internal serial transmitter of said first inboard BCC of said first brake control unit and an internal serial receiver of said first outboard BCC of said first brake control unit. 12. The aircraft braking system of any of paragraphs 8-11, wherein said first and second pedal positions are digital values and said third pedal position is an analog value. 13. The aircraft braking system of any of paragraphs 8-12, wherein said comparison module and said first module are each software modules and said second module is a hardware module. 14. The aircraft braking system of paragraph 13, wherein said second module comprises an AND gate. 15. The aircraft braking system of any of paragraphs 4-13, wherein said first inboard BCC controls operation of at least one inboard aircraft brake and said first outboard BCC controls operation of at least one outboard aircraft brake. 16. The aircraft braking system of any of paragraphs 1-15, wherein said first pedal position is a first software pedal position and said second pedal position is a second software pedal position. 17. The aircraft braking system of any of paragraphs 4-16, further comprising: a first external signal line between said first inboard BCC of said first brake control unit and said second outboard BCC of said second brake control unit, wherein said second output is provided by said second outboard BCC of said second brake control unit to said inboard BCC of said first brake control unit via said first external signal line. 18. An aircraft braking system comprising: a first pedal pair comprising a first pedal and a second pedal; a first brake control unit operatively interconnected with a first pedal position sensor for said first pedal; a second brake control unit operatively interconnected with a

second pedal position sensor for said first pedal; wherein the first brake control unit is configured to compare an output of said first pedal position sensor with an output provided by the second brake control unit based on an output of said second pedal position sensor and to preclude braking operation if the comparison reveals that the first output and second output fail to comply with a first predetermined standard. 19. A method of controlling an aircraft braking system comprising a first pedal, a first brake control unit, and a second brake control unit, said method comprising: outputting a first signal from a first pedal position sensor to said first brake control unit regarding a position of said first pedal; outputting a second signal from a second pedal position sensor to said second brake control unit regarding said position of said first pedal; said first brake control unit providing a first output that relates to a first pedal position of said first pedal; said second brake control unit providing a second output that relates to a second pedal position of said first pedal; comparing said first output with said second output; and precluding operation of said aircraft braking system in response to displacement of said first pedal and for a first condition comprising: 1) said comparing receiving said first output from said first brake control unit and receiving said second output from said second brake control unit; and 2) said comparing determining that said first output and said second output fail to comply with a first predetermined standard. 20. The method of any of paragraph 19, wherein said first brake control unit comprises a first outboard brake control card (BCC) and a first inboard BCC, wherein said second brake control unit comprises a second outboard BCC and a second inboard BCC, and wherein said first inboard BCC executes comparing step. 21. The method of paragraph 20, wherein said providing a first output is executed by said inboard BCC of said first brake control unit and said providing a second output is executed by said outboard BCC of said second brake control unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. An understanding of the present disclosure may be further facilitated by referring to the following detailed description and claims in connection with the following drawings. While the drawings illustrate various embodiments employing the principles described herein, the drawings do not limit the scope of the claims. Reference to “in accordance with various embodiments” in this Brief Description of the Drawings also applies to the corresponding discussion in the Detailed Description.

(2) FIG. 1A illustrates an exemplary aircraft having a brake assembly in accordance with various embodiments;

(3) FIG. 1B illustrates a cross-sectional view of a brake assembly in accordance with various embodiments;

(4) FIG. 2 is a schematic of an aircraft braking system in accordance with various embodiments;

(5) FIG. 3A is a functional schematic of a primary brake control unit in accordance with various embodiments;

(6) FIG. 3B is a functional schematic of a secondary brake control unit in accordance with various embodiments; and

(7) FIG. 4 is a flowchart of a method of controlling an aircraft braking system in accordance with various embodiments.

DETAILED DESCRIPTION

(8) As used herein, a first component that is “radially outward” of a second component means that the first component is positioned at a greater distance away from a common axis (e.g., a rotational axis of a wheel assembly) than the second component. A first component that is “radially inward” of a second component means that the first component is positioned closer to the common axis than

the second component. In the case of components that rotate about a common axis, a first component that is radially inward of a second component rotates through a circumferentially shorter path than the second component. As used herein, “distal” refers to the direction outward, or generally, away from a reference component. As used herein, “proximal” and/or “proximate” refer to a direction inward, or generally, towards the reference component. All ranges may include the upper and lower values, and all ranges and ratio limits disclosed herein may be combined. Unless specifically stated otherwise, reference to “a,” “an” or “the” may include one or more than one and reference to an item in the singular may also include the item in the plural.

(9) Referring to FIG. 1A, in accordance with various embodiments, an aircraft **10** is illustrated. The aircraft **10** includes landing gear, which may include a left main landing gear **12**, a right main landing gear **14** and a nose landing gear **16**. The landing gear support the aircraft **10** when it is not flying, allowing the aircraft **10** to taxi, take off and land without damage. While the disclosure refers to the three landing gear configurations just referred, the disclosure nevertheless contemplates any number of landing gear configurations.

(10) Referring now to FIG. 1B, there is schematically depicted a brake assembly or a brake mechanism **100** that may be used by the aircraft **10** of FIG. 1A or any other appropriate aircraft. The brake mechanism **100** is mounted on an axle **102** for use with a wheel **104** disposed on and configured to rotate about the axle **102** via one or more bearing assemblies **103**. The wheel **104** includes a hub **106**, a wheel well **108** concentric about the hub **106** and a web portion **110** interconnecting the hub **106** and the wheel well **108**. A central axis **112** extends through the axle **102** and defines a center of rotation of the wheel **104**. A torque plate barrel **114** (sometimes referred to as a torque tube or barrel or a torque plate or back leg) is aligned concentrically with the hub **106**, and the wheel **104** is rotatable relative to the torque plate barrel **114**.

(11) The brake mechanism **100** includes a piston assembly **116**, a pressure plate **118** disposed adjacent the piston assembly **116**, an end plate **120** positioned a distal location from the piston assembly **116**, and a plurality of rotor disks **122** interleaved with a plurality of stator disks **124** positioned intermediate the pressure plate **118** and the end plate **120**. The pressure plate **118**, the plurality of rotor disks **122**, the plurality of stator disks **124** and the end plate **120** together form a brake heat sink or brake stack **126**. The pressure plate **118**, the end plate **120** and the plurality of stator disks **124** are mounted to the torque plate barrel **114** and remain rotationally stationary relative to the axle **102**. Each stator disk **124** includes a pair of oppositely disposed sides **124a**, **124b** that each interface with a corresponding rotor disk **122** during a braking operation.

(12) The torque plate barrel **114** may include an annular barrel or torque tube **128** and an annular plate or back leg **130**. The back leg **130** is disposed at an end distal from the piston assembly **116** and may be made monolithic with the torque tube **128**, as illustrated in FIG. 1B, or may be made as a separate annular piece and suitably connected to the torque tube **128**. The torque tube **128** has a plurality of circumferentially spaced and axially extending splines **132** disposed on an outer surface of the torque tube **128**. The plurality of stator disks **124** and the pressure plate **118** include notches or stator slots **134** on an inner periphery of the disks and the plate for engagement with the splines **132**, such that each disk and the plate are axially slidable with respect to the torque tube **128**.

(13) The end plate **120** is suitably connected to the back leg **130** of the torque plate barrel **114** and is held non-rotatable, together with the plurality of stator disks **124** and the pressure plate **118**, during a braking action. The plurality of rotor disks **122**, interleaved between the pressure plate **118**, the end plate **120** and the plurality of stator disks **124**, each have a plurality of circumferentially spaced notches or rotor lugs **136** along an outer periphery of each disk for engagement with a plurality of torque bars **138** that is secured to or made monolithic with an inner periphery of the wheel **104**.

(14) An actuating mechanism for the brake mechanism **100** includes a plurality of piston assemblies, including the piston assembly **116**, circumferentially spaced around an annular piston housing **156** (only one piston assembly is illustrated in FIG. 1B). Upon actuation, the plurality of

piston assemblies affect a braking action by urging the pressure plate **118** and the plurality of stator disks **124** into frictional engagement with the plurality of rotor disks **122** and against the end plate **120**. Fluid or hydraulic pressure, mechanical springs or electric actuators, among other mechanisms, may be used to actuate the plurality of piston assemblies. Through compression of the plurality of rotor disks **122** and the plurality of stator disks **124** between the pressure plate **118** and the end plate **120**, the resulting frictional contact slows or stops or otherwise prevents rotation of the wheel **104**. The plurality of rotor disks **122** and the plurality of stator disks **124** are fabricated from various materials, such as ceramic matrix composites, that enable the brake disks to withstand and dissipate the heat generated during and following a braking action.

(15) The torque plate barrel **114** is secured to a stationary portion of the landing gear such as the axle **102**, preventing the torque plate barrel **114** and the plurality of stator disks **124** from rotating during braking of the aircraft. The torque tube **128** portion of the torque plate barrel **114** may be attached to the annular piston housing **156** via an annular mounting surface **158**, wherein bolt fasteners **160** secure the torque plate barrel **114** to the annular piston housing **156**. A spacer member or pedestal **162** is positioned between an inner diameter surface **164** of the torque tube **128** and an outer diameter surface **166** of the axle **102**. The pedestal **162** includes a radially inner surface or foot **168** for engaging the axle **102**, a web portion **170** radially outward of the foot **168** and a head portion **172** for engaging the inner diameter surface **164** of the torque tube **128**. The pedestal **162** augments support of the torque plate barrel **114** within the brake mechanism **100** generally and, more particularly, against the axle **102**. The pedestal **162** may be made monolithic with the torque tube **128** portion of the torque plate barrel **114**.

(16) A heat shield **140** is secured directly or indirectly to the wheel **104** between a radially inward surface of the wheel well **108** and the plurality of torque bars **138**. As illustrated in FIG. **1B**, the heat shield **140** is concentric with the wheel well **108** and may have a plurality of heat shield sections **142** disposed between respective, adjacent pairs of the plurality of torque bars **138**. The heat shield **140**, or heat shield sections **142**, is spaced from the radially inward surface of the wheel well **108** and secured in place by heat shield tabs **190**, such that the heat shield **140**, or heat shield sections **142**, is disposed generally parallel to the axis of rotation or central axis **112** of the wheel **104** and intermediate the plurality of torque bars **138** and the radially inward surface of the wheel well **108**. In various embodiments, including for heavy-duty applications, the heat shield **140**, or heat shield sections **142**, may be further secured in place by heat shield carriers **144**.

(17) The plurality of torque bars **138** is attached at axially inboard ends to the wheel **104** by torque bar bolts **146**. The torque bar bolts **146** extend through respective holes in a flange **150** provided on the wheel **104** as shown, which flange **150** for purposes of the present description is intended to be considered as part of the wheel well **108**. Each of the plurality of torque bars **138** may include a pin **152** or similar member at its axially outboard end (i.e., the end opposite the torque bar bolts **146**) that is received within a hole **154** disposed proximate the web portion **110** of the wheel **104**. The heat shield **140**, or heat shield sections **142**, is positioned adjacent a radially inward surface of the wheel well **108** and secured in place by the heat shield tabs **190**.

(18) An aircraft braking system is illustrated in FIG. **2** and is identified by reference numeral **200**. The aircraft braking system **200** is illustrated in conjunction with an outboard wheel **234** and an inboard wheel **238** on each of a port side **230** and a starboard side **232** of an aircraft. A separate brake **240** is provided for each of the outboard wheels **234** and the inboard wheels **238**. A hydraulic system A pressure source **250** is fluidly connected with an outboard dual brake control valve module **252** (in turn fluidly connected with the brake **240** for each of the outboard wheels **234**) and an inboard dual brake control valve module **254** (in turn fluidly connected with the brake **240** for each of the inboard wheels **238**). The aircraft braking system **200** further includes a hydraulic system A return **256**. Similarly, a hydraulic system B pressure source **260** is fluidly connected with an outboard dual brake control valve module **262** (in turn fluidly connected with the brake **240** for each of the outboard wheels **234**) and an inboard dual brake control valve module **264** (in turn

fluidly connected with the brake **240** for each of the inboard wheels **238**). The aircraft braking system **200** further includes a hydraulic system B return **266**. The aircraft braking system further includes a plurality of hydraulic fuses **270**, pressure transducers **272**, a 28 VDC (DC voltage) bus **284**, and a 28 VDC bus **286**.

(19) The above-noted brake control modules **252**, **254**, **262**, **264** may each utilize one or more servo valves, one or more shut-off valves, or the like. Each shut-off valve for each of the above-noted brake control modules **252**, **254**, **262**, **264** is disposed in the flowpath between its corresponding hydraulic system pressure source and its corresponding servo valve(s), with each shut-off valve being movable between an open and closed position in relation to the flow to its corresponding servo valve(s), and with each servo valve controlling the amount of flow to the brake of the corresponding wheel(s) (more specifically the braking pressure applied to the corresponding wheel(s)). As such, the shut-off valve(s) for each of the outboard dual brake control valve module **252** and the inboard dual brake control valve module **254** is disposed between its corresponding servo valve(s) and the hydraulic system A pressure source **250**. Similarly, the shut-off valve(s) for each of the outboard dual brake control valve module **262** and the inboard dual brake control valve module **264** is disposed between its corresponding servo valve(s) and the hydraulic system B pressure source **260**.

(20) The aircraft braking system **200** of FIG. 2 further includes a first or pilot pedal pair **210** and a second/co-pilot pedal pair **220**. The first pedal pair **210** includes a first (e.g., left) pedal **212** and a second (e.g., right) pedal **214**. A first pedal position sensor **212a** and a second pedal position sensor **212b** monitor the position of the first pedal **212** (e.g., displacement of the first pedal **212**). Similarly, a first pedal position sensor **214a** and a second pedal position sensor **214b** monitor the position of the second pedal **214** (e.g., displacement of the second pedal **214**). The second pedal pair **220** includes a first (e.g., left) pedal **222** and a second (e.g., right) pedal **224**. A first pedal position sensor **222a** and a second pedal position sensor **222b** monitor the position of the first pedal **222** (e.g., displacement of the first pedal **222**). Similarly, a first pedal position sensor **224a** and a second pedal position sensor **224b** monitor the position of the second pedal **224** (e.g., displacement of the second pedal **224**). Each of the pedal position sensors **212a/b**, **214 a/b**, **222a/b**, and **224 a/b** may be in the form of a linear variable differential transformer (LVDT).

(21) The aircraft braking system **200** utilizes both a first (e.g., primary) brake control unit (BCU) **320** and a second (e.g., secondary or alternate) BCU **330**. The first BCU **320** includes an inboard brake control card (BCC) **322** and an outboard BCC **324**. Similarly, the second BCU **330** includes an inboard BCC **332** and an outboard BCC **334**. FIG. 2 illustrates the following: 1) an output of the first pedal position sensor **212a** for the first pedal **212** (first pedal pair **210**) is provided to the outboard BCC **324** of the first BCU **320**; 2) an output of the second pedal position sensor **212b** for the first pedal **212** (first pedal pair **210**) is provided to the inboard BCC **332** of the second BCU **330**; 3) an output of the first pedal position sensor **214a** for the second pedal **214** (first pedal pair **210**) is provided to the inboard BCC **322** of the first BCU **320**; 4) an output of the second pedal position sensor **214b** for the second pedal **214** (first pedal pair **210**) is provided to the outboard BCC **334** of the second BCU **330**; 5) an output of the first pedal position sensor **222a** for the first pedal **222** (second pedal pair **220**) is provided to the outboard BCC **324** of the first BCU **320**; 6) an output of the second pedal position sensor **222b** for the first pedal **222** (second pedal pair **220**) is provided to the inboard BCC **332** of the second BCU **330**; 7) an output of the first pedal position sensor **224a** for the second pedal **224** (second pedal pair **220**) is provided to the inboard BCC **322** of the first BCU **320**; and 8) an output of the second pedal position sensor **224b** for the second pedal **224** (second pedal pair **220**) is provided to the outboard BCC **334** of the second BCU **330**.

(22) FIGS. 3A and 3B present further details regarding the first BCU **320** and the second BCU **330** from the aircraft braking system **200** of FIG. 2. The first BCU **320** again includes an inboard BCC **322** and an outboard BCC **324** (FIG. 3A), while the second BCU **330** includes an inboard BCC **332** and an outboard BCC **334** (FIG. 3B). The inboard BCC **322** and the outboard BCC **324** of the first

BCU 320 each include an external connector 290. Similarly, the inboard BCC 332 and the outboard BCC 334 of the second BCU 330 each include an external connector 290. These external connectors 290 accommodate external communications.

(23) The inboard BCC 322 of the first BCU 320 shown in FIG. 3A may be for the right pedal 214 of the first pedal pair 210 of FIG. 2 and includes the first pedal position sensor 214a (or may be for the right pedal 224 of the second pedal pair 220 of FIG. 2 in which case the inboard BCC 322 of the first BCU 320 of FIG. 3A would then include the first pedal position sensor 224a); 2) the outboard BCC 334 of the second BCU 330 shown in FIG. 3B may be for the right pedal 214 of the first pedal pair 210 of FIG. 2 and includes the second pedal position sensor 214b (or may be for the right pedal 224 of the second pedal pair 220 of FIG. 2 in which case the outboard BCC 334 of the second BCU 330 of FIG. 3B would then include the second pedal position sensor 224b); 3) the outboard BCC 324 of the first BCU 320 shown in FIG. 3A may be for the left pedal 212 of the first pedal pair 210 of FIG. 2 and includes the first pedal position sensor 212a (or may be for the left pedal 222 of the second pedal pair 220 of FIG. 2 in which case the outboard BCC 324 of the first BCU 320 of FIG. 3A would then include the first pedal position sensor 222a); and 4) the inboard BCC 332 of the second BCU 330 shown in FIG. 3B may be for the left pedal 212 of the first pedal pair 210 of FIG. 2 and includes the second pedal position sensor 214b (or may be for the left pedal 222 of the second pedal pair 220 of FIG. 2 in which case the inboard BCC 332 of the second BCU 330 of FIG. 3B would then include the second pedal position sensor 224b).

(24) Transmissions from the inboard BCC 322 of the first BCU 320 to the outboard BCC 334 of the second BCU 330 are transmitted via a signal line 340 that extends between the corresponding external connectors 290. Transmissions from the outboard BCC 334 of the second BCU 330 are transmitted to the inboard BCC 322 of the first BCU 320 via a signal line 342 that extends between the corresponding external connectors 290. Transmissions from the outboard BCC 324 of the first BCU 320 are transmitted to the inboard BCC 332 of the second BCU 330 via a signal line 344 that extends between the corresponding external connectors 290. Transmissions from the inboard BCC 332 of the second BCU 330 are transmitted to the outboard BCC 324 of the first BCU 320 via a signal line 346 that extends between the corresponding external connectors 290. Details regarding the above-noted transmissions will be discussed in more detail below.

(25) The inboard BCC 322 of the first BCU 320, the outboard BCC 324 of the first BCU 320, the inboard BCC 332 of the second BCU 330, and the outboard BCC 334 of the second BCU 330 each include the following components, and with the lines between components in FIGS. 3A and 3B being signal lines (certain outputs on one or more of the signal lines are addressed below and are identified by an additional reference numeral): a pedal position determination module 292 (e.g., a hardware pedal determination module 292); a pedal position determination module 294 (e.g., a software pedal position determination module 294); a threshold/validity module 296; an agreement/validity module 298 (or a comparison module 298); a first module 300 (e.g., providing an “AND” functionality when the first module 300 is of a software configuration); digital outputs 302/314; modules or AND gates 304/316 (when of a hardware configuration); an external serial transmitter 310a; an external serial receiver 310b; an internal serial transmitter 312a; and an internal serial receiver 312b.

(26) A representative operation of the aircraft braking system 200 will now be addressed for the right pedal 214 of the first pedal pair 210 (FIG. 2) in relation to FIGS. 3A and 3B. Operation (e.g., displacement) of the right pedal 214 is monitored by the first pedal position sensor 214a (a corresponding signal being transmitted to the external connector 290 of the inboard BCC 322 of the first BCU 320 (FIG. 3A)). The pedal position determination module 292 of the inboard BCC 322 of the first BCU 320 will determine/calculate a third pedal position (e.g., analog; a hardware pedal position) from the signal provided by the first pedal position sensor 214a. This third pedal position is assessed by the inboard BCC 322 of the first BCU 320 to determine if it satisfies a predetermined threshold (e.g., is greater than a predetermined value). If so, a sixth output 360 in the

form of a “TRUE” value/condition is transmitted to the AND gate **304** of the inboard BCC **322** of the first BCU **320**. Otherwise, a sixth output **360** in the form of a “FALSE” value/condition is transmitted to the AND gate **304** of the inboard BCC **322** of the first BCU **320**. In any case, the pedal position determination module **292** or another module of the inboard BCC **322** of the first BCU **320** could provide the assessment as to whether the third pedal position satisfies the corresponding predetermined threshold.

(27) The pedal position determination module **294** of the inboard BCC **322** of the first BCU **320** will determine/calculate a first pedal position (e.g., digital; a first software pedal position) from the signal provided by the first pedal position sensor **214a**, and this first pedal position is then transmitted to both the threshold/validity module **296** and the agreement/validity module **298** of the inboard BCC **322** of the first BCU **320** (e.g., a first output **350**). In the event that the threshold/validity module **296** of the inboard BCC **322** of the first BCU **320** determines that the first pedal position is above a predetermined value (e.g., greater than 15% activation of the right pedal **214**) and is valid, a fourth output **356** in the form of “TRUE” value/condition is transmitted from the threshold/validity module **296** to the first module **300**. Otherwise, a fourth output **356** in the form of “FALSE” value/condition is transmitted from the threshold/validity module **296** to the first module **300**.

(28) The software pedal position determination module **294** of the outboard BCC **334** of the second BCU **330** will determine/calculate a second pedal position (e.g., digital; a second software pedal position) from the signal provided by the second pedal position sensor **214b** for the first pedal **214**, and this second pedal position is then transmitted (e.g., in the form of a second output **352**) to the external serial transmitter **310a**, then to the external connector **290** of the outboard BCC **334** of the second BCU **330**, then to the external connector **290** of the inboard BCC **322** of the first BCU **20** via the signal line **340**, then to the external serial receiver **310b** of the inboard BCC **322** of the first BCU **320**, and then to the agreement/validity module **298** of the inboard BCC **322** of the first BCU **320**. If the agreement/validity module **298** determines that the first pedal position (as determined/calculated by the inboard BCC **322** of the first BCU **320**) and the second pedal position (as determined/calculated by the outboard BCC **334** of the second BCU **330**) are in agreement (e.g., within a predetermined value of each other) and are each valid, a third output **354** in the form of a “TRUE” value/condition is transmitted by the agreement/validity module **298** to the first module **300** (e.g., which provides an “AND” functionality). Otherwise, a third output **354** in the form of “FALSE” value/condition is transmitted from the agreement/validity module **298** to the first module **300**.

(29) If each input to the first module **300** is of a “TRUE” condition/value, a fifth output **358** in the form of a “TRUE” value/condition is transmitted by the first module **300** to the AND gate **304** of the inboard BCC **322** of the first BCU **320**. Otherwise, a fifth output **358** in the form of a “FALSE” value/condition is transmitted from the first module **300** to the AND gate **304** of the inboard BCC **322** of the first BCU **320**. In the event that the second BCU **330** is not operational (and thereby does not provide the second pedal position to the agreement/validity module **298** of the inboard BCC **322** for the primary BCU **320**), the agreement/validity module **298** would not send any output to the first module **300**. In this case, any “TRUE” output from the first module **300** would be based solely on the first signal value from the signal provided by the first pedal position sensor **214a** to allow for operation of the aircraft braking system **200** even with the second BCU **330** not being operational.

(30) With continued reference to the inboard BCC **322** of the first BCU **320**, if the first module **300** transmits a fifth output **358** (in the form of a “TRUE” value/condition) to the AND gate **304** and if the third pedal position (determined/calculated by the pedal position module **292**) also results in the transmission of a sixth **360** output (in the form of a “TRUE” value/condition) to the AND gate **304**, the corresponding shut-off valve driver **308** is activated. The shut-off valve driver **308** of the inboard BCC **322** of the first BCU **320** is operatively interconnected with the inboard dual brake

control valve module **254** (FIG. 2), more specifically its corresponding shut-off valve(s), and when activated disposes its corresponding shut-off valve(s) in the open position. Activating the shut-off valve driver **308** (of the inboard BCC **322** of the first BCU **320**) allows pressurized hydraulic fluid to be provided to the corresponding servo valve(s) of the inboard dual brake control valve module **254** for the aircraft braking system **200** of FIG. 2 (these servo valves being separately controlled as to the braking pressure that is ultimately applied to the corresponding brake(s) **240** of the aircraft braking system **200** (e.g., brakes **240** for the inboard wheels **238**)). Otherwise, the shut-off valve driver **308** (of the inboard BCC **322** of the first BCU **320**) is maintained in a deactivated state (e.g., such that the shut-off valve(s) for the inboard dual brake control valve module **254** is maintained in its closed position such that pressurized hydraulic fluid is not provided to the corresponding servo valve(s) of the inboard dual brake control valve module **254** for the aircraft braking system **200** of FIG. 2).

(31) The inboard BCC **322** of the first BCU **320** communicates with the outboard BCC **324** of the first BCU **320** along multiple paths regarding operation (e.g., displacement) of the right pedal **214** in accordance with the foregoing. If the third pedal position (determined/calculated by the pedal position determination module **292** of the inboard BCC **322** of the first BCU **320**) is determined to satisfy a predetermined threshold in accordance with the foregoing (e.g., is greater than a predetermined value), the above-noted sixth output **360** in the form of “TRUE” value/condition is also transmitted to the AND gate **316** of the outboard BCC **324** of the first BCU **320** via the signal line shown in FIG. 3A. Otherwise, the above-noted sixth output **360** in the form of “FALSE” value/condition is transmitted to the AND gate **316** of the outboard BCC **324** of the first BCU **320** via the signal line shown in FIG. 3A.

(32) The above-noted fifth output **358** from the first module **300** of the inboard BCC **322** of the first BCU **320** is also transmitted to the AND gate **316** of the outboard BCC **324** of the first BCU **320** (via the internal serial transmitter **312a** of the inboard BCC **322** of the first BCU **320** and the internal serial receiver **310b** of the outboard BCC **324** of the first BCU **320**). Receipt of both “TRUE” outputs **358**, **360** by the AND gate **316** of the outboard BCC **324** of the first BCU **320** results in the corresponding shut-off valve driver **308** being activated. The shut-off valve driver **308** of the outboard BCC **324** of the first BCU **320** is operatively interconnected with the outboard dual brake control valve module **252** (FIG. 2), more specifically its corresponding shut-off valve(s), and when activated disposes its corresponding shut-off valve(s) in the open position. Activating the shut-off valve driver **308** (of the outboard BCC **324** of the first BCU **320**) allows pressurized hydraulic fluid to be provided to the corresponding servo valve(s) of the outboard dual brake control valve module **252** for the aircraft braking system **200** of FIG. 2 (these servo valves being separately controlled as to the braking pressure that is ultimately applied to the corresponding brake(s) **240** of the aircraft braking system **200** (e.g., brakes **240** for the outboard wheels **234**)). Otherwise, the shut-off valve driver **308** (of the outboard BCC **324** of the first BCU **320**) is maintained in a deactivated state, for instance if at least one input in the form of a “FALSE” value/condition is received by the AND gate **316** (e.g., such that the shut-off valve(s) for the outboard dual brake control valve module **252** is maintained in its closed position such that pressurized hydraulic fluid is not provided to the corresponding servo valve(s) of the outboard dual brake control valve module **252** for the aircraft braking system **200** of FIG. 2).

(33) For the case of operation (e.g., displacement) of the left pedal **212** of the first pedal pair **212**, the operations described above by the first BCU **320** would be reversed between its inboard BCC **322** and its outboard BCC **324** (and using the pedal position sensors **212a/212b**). It should also be appreciated that the foregoing is equally applicable to operation of the aircraft braking system **200** using the second pedal pair **220** and the corresponding components (e.g., where the shut-off valve driver **308** of the inboard BCC **332** of the second BCU **330** is operatively interconnected with the inboard dual brake control valve module **264**, more specifically its corresponding shut-off valve(s); where the shut-off valve driver **308** of the outboard BCC **334** of the second BCU **330** is operatively

interconnected with the outboard dual brake control valve module **262** (FIG. 2), more specifically its corresponding shut-off valve(s)).

(34) The pedal position determination module **294**, threshold/validity module **296**, agreement/validity module (or comparison module) **298**, and first module **300** each may be in software (although the noted function(s) could be provided in hardware). The pedal position determination module **292**, the AND gate **304**, and the AND date **316** may be in hardware (although the noted function(s) could be provided in software).

(35) A method of controlling operation of an aircraft braking system is illustrated in FIG. 4 and is identified by reference numeral **300**. A first signal from a first pedal sensor is output to a first brake control unit, where this first signal pertains to displacement of a first brake pedal (**302**). A second signal from a second pedal sensor is output to a second brake control unit, where this second signal also pertains to the displacement of the same first brake pedal (**304**). A first pedal position (e.g., a first software pedal position) is calculated/determined using the first brake control unit (**306**). A second pedal position (e.g., a second software pedal position) is calculated/determined using the second brake control unit (**308**). The first pedal position and the second pedal position are compared (**310**). Operation of the aircraft brake system is precluded (e.g., opening of a shut-off valve to provide pressurized fluid for operation of one or more aircraft brakes may not be allowed) in response to the existence of a first condition (**312**), where this first condition may include a comparison module receiving the first output from the first brake control unit, the comparison module receiving the second output from the second brake control unit, and the comparison module determining that the first output and the second output fail to comply with a first predetermined standard (e.g., the first output and the second output are not within a predetermined tolerance of one another).

(36) The foregoing description of the present invention has been presented for purposes of illustration and description. Furthermore, the description is not intended to limit the invention to the form disclosed herein. Consequently, variations and modifications commensurate with the above teachings, and skill and knowledge of the relevant art, are within the scope of the present invention. The embodiments described hereinabove are further intended to explain best modes known of practicing the invention and to enable others skilled in the art to utilize the invention in such, or other embodiments and with various modifications required by the particular application(s) or use(s) of the present invention. It is intended that the appended claims be construed to include alternative embodiments to the extent permitted by the prior art.

(37) Any feature of any other various aspects addressed in this disclosure that is intended to be limited to a “singular” context or the like will be clearly set forth herein by terms such as “only,” “single,” “limited to,” or the like. Merely introducing a feature in accordance with commonly accepted antecedent basis practice does not limit the corresponding feature to the singular. Moreover, any failure to use phrases such as “at least one” also does not limit the corresponding feature to the singular. Use of the phrase “at least substantially,” “at least generally,” or the like in relation to a particular feature encompasses the corresponding characteristic and insubstantial variations thereof (e.g., indicating that a surface is at least substantially or at least generally flat encompasses the surface actually being flat and insubstantial variations thereof). Finally, a reference of a feature in conjunction with the phrase “in one embodiment” does not limit the use of the feature to a single embodiment.

(38) The foregoing description has been presented for purposes of illustration and description. Furthermore, the description is not intended to limit the invention to the form disclosed herein. Consequently, variations and modifications commensurate with the above teachings, and skill and knowledge of the relevant art, are within the scope of the present disclosure. Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the

various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Different cross-hatching is used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

(39) Systems, methods and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “various embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(40) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Finally, it should be understood that any of the above described concepts can be used alone or in combination with any or all of the other above described concepts. Although various embodiments have been disclosed and described, one of ordinary skill in this art would recognize that certain modifications would come within the scope of this disclosure. Accordingly, the description is not intended to be exhaustive or to limit the principles described or illustrated herein to any precise form. Many modifications and variations are possible in light of the above teaching.

Claims

1. An aircraft braking system comprising: a first pedal pair comprising a first pedal and a second pedal; a first pedal position sensor for said first pedal; a second pedal position sensor for said first pedal; a first brake control unit; a second brake control unit, wherein said first pedal position sensor is operatively interconnected with one of said first brake control unit and said second brake control unit and said second pedal position sensor is operatively interconnected with the other of said first brake control unit and said second brake control unit; and a first comparison module configured to compare a first output with a second output, wherein said first output is provided by said first brake control unit from an output of said first pedal position sensor and relates to a first pedal position of said first pedal, wherein said second output is provided by said second brake control unit from an output of said second pedal position sensor and relates to a second pedal position, and wherein said

first brake control unit comprises said first comparison module; wherein a first condition precludes operation of said aircraft braking system in response to displacement of said first pedal, said first condition comprising: 1) said first comparison module receiving said first output from said first brake control unit and receiving said second output from said second brake control unit; and 2) said first comparison module determining that said first output and said second output fail to comply with a first predetermined standard.

2. The aircraft braking system of claim 1, wherein said first output comprises both a validity and magnitude relating to said first pedal position, and wherein said second output comprises both a validity and magnitude of said second pedal position.

3. The aircraft braking system of claim 1, wherein one of said first brake control unit and said second brake control unit is a primary brake control unit and the other of said first brake control unit and said second brake control unit is a secondary brake control unit.

4. The aircraft braking system of claim 1, wherein said first brake control unit comprises a first outboard brake control card (BCC) and a first inboard BCC, wherein said second brake control unit comprises a second outboard BCC and a second inboard BCC, and wherein said first inboard BCC of said first brake control unit comprises said first comparison module.

5. The aircraft braking system of claim 4, wherein said first pedal position sensor is operatively interconnected with said first inboard BCC of said first brake control unit and said second pedal position sensor is operatively interconnected with said second outboard BCC of said second brake control unit.

6. The aircraft braking system of claim 5, wherein said first output is provided by said first inboard BCC of said first brake control unit and said second output is provided by said second outboard BCC of said second brake control unit.

7. The aircraft braking system of claim 5: wherein said first inboard BCC of said first brake control unit comprises said first comparison module, a first module, and a second module; wherein said first module utilizes a third output from said first comparison module and a fourth output, wherein said fourth output is provided by said first brake control unit from an output of said first pedal position sensor and relates to said first pedal position of said first pedal; and wherein said first module outputs a fifth output to said second module.

8. The aircraft braking system of claim 7, wherein a sixth output is provided by said first brake control unit to said second module from an output of said first pedal position sensor and that relates to a third pedal position of said first pedal, and wherein said second module is operatively interconnected with a first shut-off valve driver that opens a first shut-off valve of said aircraft braking system when said fifth output satisfies a first condition and said sixth output satisfies a second condition.

9. The aircraft braking system of claim 8, wherein satisfaction of said first condition at least requires said first pedal position to be both above a first predetermined threshold and to be valid, and wherein satisfaction of said second condition at least requires said third pedal position to be above a second predetermined threshold.

10. The aircraft braking system of claim 8: wherein said first outboard BCC of said first brake control unit comprises a third module; wherein said first module also transmits said fifth output to said third module; wherein said first inboard BCC of said first brake control unit also transmits said sixth output to said third module; and wherein said third module is operatively interconnected with a second shut-off valve that opens a second shut-off valve of said aircraft braking system when said fifth output satisfies said first condition and said sixth output satisfies said second condition.

11. The aircraft braking system of claim 10, wherein said first module transmits said fifth output to said third module via an internal serial transmitter of said first inboard BCC of said first brake control unit and an internal serial receiver of said first outboard BCC of said first brake control unit.

12. The aircraft braking system of claim 8, wherein said first and second pedal positions are digital values and said third pedal position is an analog value.
13. The aircraft braking system of claim 7, wherein said comparison module and said first module are each software modules and said second module is a hardware module.
14. The aircraft braking system of claim 13, wherein said second module comprises an AND gate.
15. The aircraft braking system of claim 4, wherein said first inboard BCC controls operation of at least one inboard aircraft brake and said first outboard BCC controls operation of at least one outboard aircraft brake.
16. The aircraft braking system of claim 4, further comprising: a first external signal line between said first inboard BCC of said first brake control unit and said second outboard BCC of said second brake control unit, wherein said second output is provided by said second outboard BCC of said second brake control unit to said inboard BCC of said first brake control unit via said first external signal line.
17. An aircraft braking system comprising: a first pedal pair comprising a first pedal and a second pedal; a first brake control unit operatively interconnected with a first pedal position sensor for said first pedal; a second brake control unit operatively interconnected with a second pedal position sensor for said first pedal; wherein the first brake control unit is configured to compare an output of said first pedal position sensor with an output provided by the second brake control unit based on an output of said second pedal position sensor and to preclude braking operation if the comparison reveals that the first output and second output fail to comply with a first predetermined standard.
18. A method of controlling an aircraft braking system comprising a first pedal, a first brake control unit, and a second brake control unit, said method comprising: outputting a first signal from a first pedal position sensor to said first brake control unit regarding a position of said first pedal; outputting a second signal from a second pedal position sensor to said second brake control unit regarding said position of said first pedal; said first brake control unit providing a first output that relates to a first pedal position of said first pedal; said second brake control unit providing a second output that relates to a second pedal position of said first pedal; comparing said first output with said second output; and precluding operation of said aircraft braking system in response to displacement of said first pedal and for a first condition comprising: 1) said comparing receiving said first output from said first brake control unit and receiving said second output from said second brake control unit; and 2) said comparing determining that said first output and said second output fail to comply with a first predetermined standard.
19. The method of any of claim 18, wherein said first brake control unit comprises a first outboard brake control card (BCC) and a first inboard BCC, wherein said second brake control unit comprises a second outboard BCC and a second inboard BCC, and wherein said first inboard BCC executes the comparing step.
20. The method of claim 19, wherein said providing a first output is executed by said inboard BCC of said first brake control unit and said providing a second output is executed by said outboard BCC of said second brake control unit.
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